

JINQI GONG

☎ (510)256-9569 ✉ eggmangong@gmail.com 🔗 linkedin.com/in/gongjinqi 🐙 eggmangong.github.io

EDUCATION

University of Southern California

California, USA

Master of Science in Analytics | Industrial & Systems Engineering, Viterbi School of Engineering 08/2024 - 05/2026 (Exp.)
Coursework: Predictive Analytics, Optimization Method, Database Management, Business Intelligence GPA: 4.0/4.0

University of Macau

Macau, China

Bachelor of Science in Mathematics | Statistics & Data Science, Faculty of Science and Technology 09/2020 - 06/2024
Coursework: Linear Algebra, Programming, Advanced Probability, Statistical Analysis, Machine Learning GPA: 3.66/4.0

EXPERIENCE

Grader/Course Producer, University of Southern California

01/2025 - present

- For ISE530 Optimization Methods for Analytics, by Prof. Niloufar Izadinia
- Grade 60+ assignments per week with attention to accuracy and timely feedback

Viterbi Admission & Student Engagement Intern, University of Southern California

12/2024 - present

- Assist with presentations and events, and serve as representatives and liaisons for 800+ Viterbi students.
- Share photos and live stories to short blog posts weekly as an ambassador and lead campus tours

Research Assistant, Tsinghua University

06/2023 - 10/2023

- Information Retrieval Lab at Tsinghua University (THUIR) | Advisor: Shaoping Ma, Professor
- Research Area: Information Retrieval, Recommendation System, Natural Language Processing, Large Language Model
- Rewrote 300+ prompts and compared 11 Large Language Models by calling APIs for LLM evaluations

Visiting Student, University of California, Berkeley

01/2023 - 05/2023

- Spring 2023, Berkeley Global Access (BGA)
- Coursework: Artificial Intelligence, Abstract Algebra, Numerical Analysis

RESEARCH/PROJECTS

Stock Model Analysis and Investment Strategy | Python, Data Analysis, Visualization, Machine Learning

11/2023

- Silver Medal, First Prize (Top 0.17%) | "Greater Bay Area Cup" Financial Mathematical Modeling Competition
- Constructed 5 key characteristic indicators for CCV stocks, and summarize into 3 clusters by Hierarchical Clustering method
- Developed short/long-term investment portfolios that generated 10%+ returns by Markowitz Model and linear programming

Double Correction Framework for Denoising Recommendation | Pytorch, Deep learning

06/2023 - 10/2023

- Co-author | Paper accepted by the ACM SIGKDD Conference 2024
- Improved model performance by up to 25.9% through stable loss calculation and cautious hard sample search
- Designed a progressive label correction strategy, leading to an 8.6% improvement in recommendation performance

Data Augmentation and Parameter Expansion in MCMC | R, Python, Statistics, Probability

01/2024 - 05/2024

- Faculty of Science and Technology Outstanding Final Year Project (Top 3%) | Advisor: Zhi Liu, Professor
- Increased convergence speed by 30% using the Parameter Expansion Data Augmentation (PX-DA) algorithm
- Enhanced Bayesian curve fitting accuracy by 20%, leveraging data augmentation to optimize parameter estimation

Matrix Functions with Chebyshev Polynomials | MATLAB, Statistics, Probability

04/2023 - 05/2023

- Accelerated matrix function computation using Chebyshev polynomial approximation with computational efficiency
- Maintained relative error under 10^{-8} as matrix size increases, showing numerical stability in approximating matrix functions
- Achieved complexity $O(N^2)$ for large-scale matrix operations, closely matching ideal theoretical expectations

Movie Industry Database Design | SQL, Data Modeling, Database Management

04/2022 - 05/2022

- Developed a structured database system using MySQL Workbench, covering 9 entities and relations
- Optimized query efficiency by implementing 10+ SQL queries involving JOINs, aggregations, and subqueries
- Created 1 view, 2 functions and 1 procedure to enhance database usability, allowing quick retrieval

SKILLS

Programming Languages: Python, SQL, R, MATLAB, Java, C, AMPL

Tools: Tableau, PowerBI, LaTeX, Markdown, Jupyter Notebook, VS Code, Git, Excel, PowerPoint, Word

Languages: Mandarin (native), English (full professional), Cantonese (professional working)

AWARDS

David and Barbara Pong Founding Master's Scholarship (Top 1%)

2023 - 2024

Lou Tou Vo Mathematics Scholarship (Top 3%)

2020 - 2021 / 2021 - 2022

Residential College Foundation Scholarship (Top 3%)

2020 - 2021